

Revision exercises: Unit 1 – Magnetism

1. Give one term for each description.

- A region in space where another magnet or ferromagnetic material will experience a force
- Domains are lined up to create a net magnetic field in these materials
- The northernmost point of Earth's axis of rotation
- A magnetic needle that is free to turn
- Coloured lights in the sky created by the interaction of Earth's magnetic field and charged particles in the solar wind

2. Study the group of bar magnets shown below.



The following is known:

- Pole B repels pole C.
- Pole D attracts pole E.
- Pole F is a north pole.

- Write down the general rule for forces (attractive and repulsive) between magnetic poles.
- Will the magnetic force between the following poles be attractive or repulsive?
 - B and E
 - A and D
 - B and F

For questions c) and d) below: Label the magnetic poles according to the information given at the beginning of the question.

- Draw the magnetic field around magnet AB. Indicate the direction of the field.



- Draw the resultant magnetic field around the magnetic poles of D and E. Also indicate the direction of the magnetic field.



- Is the force between magnets a field or contact force. Explain your answer.
- A magnet is cut in three pieces, as shown in the sketch. Indicate the polarity of the different pieces.



Revision exercises: Unit 2 – Electrostatics

1. Give one term for each description.

- a) An object with an equal number of electrons and protons
- b) The exact charge on an electron
- c) The force acting on two bodies that carry like charges
- d) Objects that are electron deficient
- e) A region in which an electric charge experiences a force
- f) The principle that every charge in the universe consists of integer multiples of the electron charge
- g) The net charge of an isolated system remains constant during any physical process

2. A balloon is rubbed on Priya's hair and then hung from the ceiling. A negatively charged rod is brought close to the balloon and the rod repels the balloon.



- a) What charge does the balloon have?
- b) Explain how you arrive at your answer.
- c) What is this method of charging the balloon called?
- d) What is the charge on Priya's hair immediately after the balloon was rubbed on it?
- e) If 300 electrons were transferred between the balloon and the hair, what is the charge on the balloon and hair respectively? Assume that both the balloon and hair were initially uncharged.

3. Every charge in the universe consists of integer multiples of the electron charge.

- a) What is this principle called?

b) A charged object carries a charge of $-8 \times 10^{-17} \text{ C}$.

i) How many excess electrons are present on the object?

ii) If there are $4,8 \times 10^{-17} \text{ C}$ of positive charge on the object, how many electrons are present in total?

c) A charged object has a shortage of 235 electrons. What is the charge on the object?

d) Two identical metal spheres on isolated stands carry charges as follows: Sphere A has a charge of $-5,6 \times 10^{-17} \text{ C}$ and sphere B has 600 more protons than electrons.

i) What is the charge on sphere B?

ii) If the spheres are allowed to make contact and then separated again, what is the new charge on each sphere?

iii) How many electrons were transferred?

e) Two identical metal spheres, P and Q, on isolated stands carry $+2,4 \times 10^{-17} \text{ C}$ and $+3,2 \times 10^{-17} \text{ C}$, respectively. The two spheres make contact and electrons are transferred from one sphere to the other.

i) Will the transfer of electrons be from P to Q or Q to P?

ii) Determine the amount of charge that is transferred.

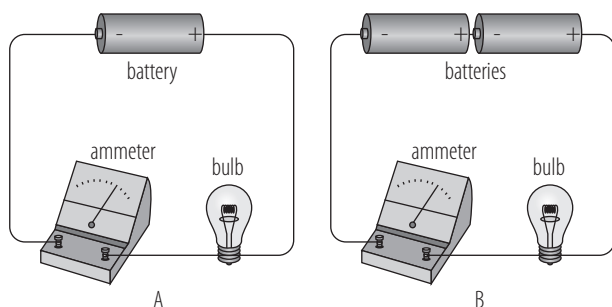
iii) How many electrons are transferred?

Revision exercises: Unit 3 – Electric circuits

1. Give one term for the following descriptions.

- Charge carriers in a metal wire
- Unit for electric energy
- Potential difference across the terminals of a battery when no current is flowing
- Resistors connected like this act as current dividers
- Resistors connected like this act as voltage dividers
- The rate of flow of charge
- Energy transferred per unit charge
- One volt per ampere

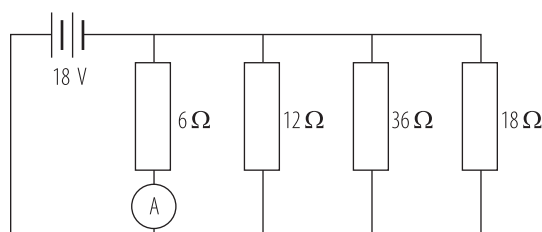
2. Consider circuits A and B. Assume that all cells and bulbs are identical.



- Explain the difference in brightness of the bulbs in circuits A and B.
- The ammeter in each circuit measures electric current.
 - Explain the meaning of conventional current.
 - Define electric current.
- The reading on the ammeter in circuit A is 0,5 A.
 - Use the definition in b) ii) to explain this value.
 - How will the current be influenced if a second light bulb is connected in series in circuit A?
- The cells in both circuits will eventually go flat.
 - Explain this statement by referring to energy conversions.
 - What do we call a battery that can be reused after it has gone flat?

iii) Give one example where such batteries are used.

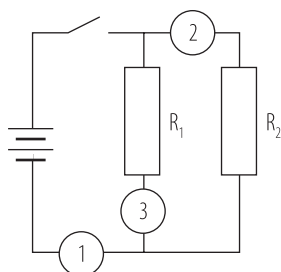
3. The diagram below represents an electric circuit consisting of four resistors and an 18-volt battery.



- Determine the equivalent resistance in the circuit.
- If the reading on the ammeter is 3 A, determine the current in the following resistors:
 - 12 Ω
 - 36 Ω
 - 18 Ω.
- What is the total current in the circuit?
- If the 36 Ω is removed from the circuit, what will the affect be on:
 - total resistance?
 - total current?
- Calculate the amount of charge that flows through the battery in 2 minutes.

4. Refer to the circuit diagram below to answer the questions.

When the switch is open (like in the diagram), the voltmeter reading across the battery is 3 V, but when the switch is closed, the reading drops to 2,8 V. Different positions (1–3) for ammeters are shown on the diagram.



- What is the emf of the battery?
- Determine the terminal voltage.
- If 7 nC of electric charge move through the battery, how much energy is transferred to the charges?
- If an ammeter is placed in positions 1 to 3, through which component(s) in the circuit will it measure the current in:
 - position 1?
 - position 2?
 - position 3?
- If R_2 has a greater resistance than R_1 , how will the potential difference across R_2 differ from that of R_1 ?